

Nicholas Wen

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Summary

I am an aspiring software developer, who is deeply interested in software and how it work under the hood.
My interests include web, mobile development, systems design, artificial intelligence as well as operating systems and compilers.

Education

Nanyang Polytechnic Singapore
Diploma in Information Technology Apr 2023 – Apr 2026

Holy Innocents' High School Singapore
in GCE 'O' Levels Jan 2018 – Dec 2022

Skills

Languages: Python, Go, Rust, C, Ada/SPARK, C, Kotlin, C#, JavaScript (TypeScript, Node, React)

Machine Learning: PyTorch, TensorFlow, LangChain, HuggingFace, ONNX

Infrastructure: Docker, Docker Compose, MySQL, MS SQL, SQLite, Redis, KeyDB

Others: Git, Bash, PowerShell, HTML, CSS, SASS, Tailwind, Bootstrap

Experience

EUC Systems Intern Singapore
Fujitsu Mar 2025 – May 2025

Assisted end users with *troubleshooting software issues*, wrote Python scripts to **automate email notifications**.
- Imaging, setting up and wiping data for **80+ Windows 11 laptops**. Tagged and accounted for over **120+ assets**.

Relevant Coursework

Automo Nanyang Polytechnic
SARIMA time-series dashboard with Docker Compose in *decoupled services architecture*. Jan 2026 – Feb 2026
- KeyDB for **in-memory caching**, Goatcounter for *observability* and Gemini **generated Executive Summary**.

NYP-FYP CNC Chatbot Nanyang Polytechnic
Refactored and decoupled an inherited chatbot platform into an **optimised, containerised** runtime architecture. June 2025 – Aug 2025
- Implemented limiting and mitigated resource exhaustion, **reducing initialisation latency by over 20 seconds**.

Projects

Ada CRDT June 2026 – Present
First Conflict-Free Replicated Data Types (CRDT) library for Ada/SPARK, achieving strong eventual consistency across decentralized nodes.
- CRDT primitives like **PN-Counters**, **LWW-Sets**, and **RGA sequences** with interchangeable backends (**Yjs**, **Fugue**, **Naive**).
- **Zero-heap bounded types**, compressed **LEB128** binary wire protocol with **Hybrid Logical Clocks (HLC)** to minimise bandwidth during partition healing.
- Real-time **Conway's Game of Life TUI demo**, supported by extensive test suite and **docs-as-code pipeline**.

yams Apr 2026 – Present
Media Screensaver Terminal User Interface (TUI) written with C and ncurses.
- Event-driven architecture via **D-Bus/MPRIS** with FFmpeg/Chafa API handling and a **200ms debounce** for snappy user experience.
- Engineered resilient asset pipeline with **local image caching** and exponential backoff retry for robust remote asset handling.
- Statically-linked **~1.5 MB binary** with **zero-leak memory profile** (Valgrind-verified) and negligible idle CPU overhead (**<2%**).

flexcyon Sept 2024 – Present
Obsidian.md theme with terminal inspired aesthetics, over **3000 downloads** and **300+ customisation options**.
- Use of SCSS for *codebase modularity*, with SemVer and linting. **Documentation** in both *English and Chinese*.

Co-curriculars

NDP 2024 Motivators May 2024 – Aug 2024
Participated in Singapore's National Day Parade.

Project Mid-Autumn Festival July 2023 – Sept 2023
Involved as **Student Organiser**, planning logistics and activities during the event at the elderly home.

HIHS Chinese Weiqi Club Feb 2021 – Feb 2022
Involved as **Executive Committee** Member, with roles in Publicity, Mentorship and organising the inaugural Weiqi Immersion Fortnight.

Awards and Honors

MOE Edusave Award for Achievement, Good Leadership and Service (EAGLES) 2021
Ministry of Education (MOE)

Certifications

Certificate of Participation for Beyond Borders with DDAS | Hong Kong Edition 2025
Digital Defence Alliance Singapore (DDAS)

Google Data Analytics 2025
Coursera

Google IT Automation with Python 2024
Coursera